To get a basic grasp of this process for CoffeeScript, consider the simple part of the CoffeeScript syntax that Scruffy wants to change: function syntax. The basic expression that you'll examine in the CoffeeScript compiler is the identity function assigned to a variable `I`:

```
I = (x) -> x
```

The first thing the compiler must do is convert this string (input stream) to tokens.

Compiler versions

The CoffeeScript compiler version described here is CoffeeScript 1.6.2, so depending on the version of the compiler you have, the results you get when inspecting the compiler may be different (particularly regarding the things that get rewritten). But the general structure of the compiler will be the same, regardless of the version you're using.

8.3.1. Tokenizing

During compilation, the first step is to use a *lexer* to covert the input into an array of tokens known as the token stream. Consider the tokens that the identity function assignment expression produces: